

Upsampling and Downsampling: Correcting the Imbalances in Data

By mastering the art of upsampling and downsampling, we not only correct skewed datasets but also pave the way for fairer, more accurate machine learning models. This second article in the two-part series focuses on the uses of downsampling.



In the first part of this article series, which was published in the March 2024 edition of *OSFY*, we learnt how to manage imbalanced datasets by using upsampling techniques. Now, let us see how downsampling works.

Downsampling

Downsampling is used to decrease the size of a dataset by removing some of the instances. When the data is imbalanced, the data points of the majority class are removed. This is often done to reduce computational complexity and training time, and to eliminate redundancy or irrelevant instances from the dataset. However, when doing this, we need to ensure important information is not lost and the sample represents the entire dataset.

Methods used in downsampling

Upsampling is used when the dataset is large. Downsampling is a good option to reduce the computational complexity and training time of the model. If the goal is to improve model efficiency or reduce the risk of overfitting, downsampling is a better option. A few methods used for downsampling are explained here, though these are not exhaustive.

Resample

- This method which is used for upsampling can be used for downsampling as well. The simplest way of downsampling majority classes is by randomly removing records from that category.
- Logic used is the same as upsampling — instead of increasing random records, remove them.
- Here is sample code:

```
from collections import Counter
from sklearn.datasets import make_classification
from sklearn.utils import resample
from matplotlib import pyplot
from numpy import where
import pandas as pd
import numpy as np
```

```
X, y = make_classification(n_classes=2, class_sep=2,
weights=[0.1, 0.9], n_features=4, n_clusters_per_class=1,
n_samples=1000, random_state=10)
```

```
# summarize class distribution
```